Jan Kaiser

RESEARCH SCIENTIST IN ARTIFICIAL INTELLIGENCE FOR PARTICLE ACCELERATORS

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BRIEF PROFILE

Postdoctoral research scientist at the intersection of artificial intelligence, control theory and physics; building solutions to autonomously operate particle accelerators with the help of reinforcement learning and differentiable simulations. As part of my postdoctoral and PhD work at DESY I develop solutions to overcome challenges of deploying reinforcement learning to real-world systems, such as sample-efficiency and sim-to-real transfer. The developed reinforcement learning agents have been deployed to accelerator operations and outperformed expert human operators. I further manage the open source high-speed differentiable simulator *Cheetah*, which I developed during my PhD, and which has since grown to be the foremost differentiable simulator for particle accelerators. My other research interests include surrogate modelling and digital twins using a variety of supervised learning methods. My work has been published at top-tier journals and conferences, including ICML and Science Advances. I further have a solid background in Computer Science, in which I completed my Master's and Bachelor's degrees, researching anomaly detection for unmanned aerial vehicles using recurrent neural networks.

PROFESSIONAL EXPERIENCE

Deutsches Elektronen-Synchrotron DESY, Hamburg
Research Scientist

SEPTEMBER 2024 - PRESENT

- Develop reinforcement learning-based solutions for more complex and high impact tasks where training sample generation in simulation is also infeasibly expensive (e.g. autonomously operating parts of the *European XFEL* – one of the largest particle accelerators in the world).
- Apply novel solutions, such as gradient-based reinforcement learning and vectorised simulations, enabled by a novel differentiable simulator I developed.
- Continued development and strategic management of the globally used open source project *Cheetah* (a high-speed differentiable beam dynamics simulator) with 20 contributors and over 1,000 monthly downloads.
- Act as deputy leader of the *Intelligent Process Controls* team responsible for forward developments of advanced control schemes and anomaly detection methods for particle accelerators.
- Organise annual scientific workshops as part of the *Reinforcement Learning for Autonomous Accelerators (RL4AA)* collaboration – an international effort annually bringing together up to 90 researchers interested in applying reinforcement learning to particle accelerators and other real-world systems.
- Collaborate in interdisciplinary teams with physicists, control engineers and data scientists.
- Publish and present research results at international conferences and in peer-reviewed journals.
- Supervise students' Master's theses, Bachelor's theses, and research projects.

Deutsches Elektronen-Synchrotron DESY, Hamburg
Doctoral Researcher

DECEMBER 2020 - AUGUST 2024

- Research topics: reinforcement learning, differentiable simulations and digital twins for particle accelerators.
- Develop novel reinforcement solutions for particle accelerator tuning and control, solving challenges like the high cost of training samples, sim-to-real transfer, partial observability and noisy real-world observations, ultimately achieving zero-shot application to the real world with better performance than expert human operators.

- Build the beam dynamics simulator Cheetah based on PyTorch, offering orders of magnitude faster and vectorised simulations, built-in gradient computation, and native integration with machine learning software. Apply Cheetah to tasks including accelerator tuning, system identification, Bayesian optimisation with informed priors, and AI/ML coupling.
- Investigate the use of large language models for numerical black-box optimisation and particle accelerator tuning.
- Train neural network-based digital twins for online virtual diagnostics.
- Share scientific results through journal and conference papers, presentations, posters, and tutorials.
- Supervise students' Master's theses, Bachelor's theses, and research projects.

Hamburg University of Technology, Hamburg
Teaching Assistant

NOVEMBER 2016 - JANUARY 2020

- Teach and supervise lab and exercise sessions, as well as grade student works in multiple different courses on different subjects.
- Courses taught: Programming in C; Object Oriented Programming, Algorithms, and Data Structures; Numerics and MATLAB

Marquard & Bahls AG, Hamburg
Intern

JULY 2015 - SEPTEMBER 2015

- Observe and assist in various tasks of the IT department in a large international energy cooperation, on both the software development and IT operations sides.
- Follow meetings on the development of custom operations software, as well as meetings decision making on software purchasing evaluations.
- Develop a web-based system status notification ticker for the IT helpdesk.

Nähmaschinenfabrik Emil Stutznäcker GmbH & Co. KG, Köln
Intern

JULY 2014 - AUGUST 2014

- Observe and assist in different stages of the manufacturing of industrial sawing machines, including assembly and the operation of large industrial CNC machines.
- Learn various techniques for manufacturing custom metal parts by hand.

EDUCATION

Hamburg University of Technology, Hamburg
Doctor of Philosophy (PhD) Computer Science

MARCH 2021 - JUNE 2025

Supervisor: Prof. Dr.-Ing. Annika Eichler (DESY / TUHH, Institute for Control Systems)
2nd examiner: Prof. Dr.-Ing. Görschwin Fey (TUHH, Institute for Embedded Systems)
Title: Reinforcement Learning and Differentiable Simulations for Autonomous Tuning and Control of Linear Particle Accelerators
Final grade: "with distinction" (summa cum laude)

Hamburg University of Technology, Hamburg
Master of Science (M. Sc.) Computer Science

SEPTEMBER 2018 - OCTOBER 2020

Final grade: 1.4
Thesis: Monitoring Embedded Devices in the Loop using Recurrent Neural Networks

Hamburg University of Technology, Hamburg
Bachelor of Science (B. Sc.) Computer Science

OCTOBER 2015 - AUGUST 2018

Final grade: 2.0
Thesis: Trajectory Anonymisation by Multivariate Microaggregation

PROJECTS & COLLABORATIONS

Cheetah Differentiable Beam Dynamics Simulator
Maintainer and original developer

2021 - PRESENT

Cheetah is a high-speed fully-vectorised and differentiable beam dynamics simulator built on top of PyTorch to support various applications, such as reinforcement learning, accelerator tuning, system identification, Bayesian optimisation with informed priors, and AI/ML coupling. Development was originally started by me to solve the cost of collecting training samples for reinforcement learning. Since then, Cheetah has become one of the fastest beam dynamics simulators available, the leading differentiable beam dynamics simulator, and an international open source project with over 25000 downloads and 20 contributors from 11 institutes.

URL: https://github.com/desy-ml/cheetah

Reinforcement Learning for Autonomous Particle Accelerators Collaboration (RL4AA)
Co-founder and member of the steering committee

2023 - PRESENT

I co-founded RL4AA, an international collaboration with the goal of fostering international collaboration and knowledge exchange on the application of reinforcement learning to particle accelerators. As a member of the RL4AA steering committee, I have helped organise annual workshops with around 90 participants, and secure world-class keynote speakers for these events.

Link: https://rl4aa.github.io/

AWARDS & SCHOLARSHIPS

Physical Review Accelerators and Beams Top 10 Highlight Paper Hand-picked by The Editors from *American Physical Society*. 2024.

Kongressreisen 2024 Scholarship from *German Academic Exchange Service (DAAD)*. 2024.

TEACHING & SUPERVISION

Master's theses: 2 | *Bachelor's theses: 2* | *Research projects: 2* | *Student assistants & interns: 5*

SELECTED PUBLICATIONS

Citations: 360 | *h-index: 10* | *i10-index: 10 – (Google Scholar 03/2026)*

C. Hespe, J. Kaiser, J. O. Lübsen, F. Mayet, M. Scholz and A. Eichler. "**Data-driven Feedback Optimization for Particle Accelerator Application**," in *at - Automatisierungstechnik*, 2025.

A. Santamaria Garcia, C. Xu, A. Eichler, J. Kaiser and S. Hirlaender. "**Efficient Data-driven Model Predictive Control for Online Accelerator Tuning**," in *16th International Particle Accelerator Conference (IPAC)*, 2025.

J. Kaiser, A. Lauscher and A. Eichler. "**Large Language Models for Human-Machine Collaborative Particle Accelerator Tuning through Natural Language**," in *Science Advances*, 2025.

R. Roussel, G. Charleux, A. L. Edelen, A. Eichler, J. P. Gonzalez-Aguilera, A. Huebl, J. Kaiser, R. Lehe, A. Santamaria Garcia and C. Xu. "**Advancements in Backwards Differentiable Beam Dynamics Simulations for Accelerator Design, Model Calibration, and Machine Learning**," in *32nd Linear Accelerator Conference (LINAC2024)*, 2024.

S. Hirlaender, S. Pochaba, L. Lamminger, A. Santamaria Garcia, C. Xu, J. Kaiser, A. Eichler and V. Kain. "**Deep Meta Reinforcement Learning for Rapid Adaptation In Linear Markov Decision Processes: Applications to CERN's AWAKE Project**," in *Combining, Modelling and Analyzing Imprecision, Randomness and Dependence. SMPS 2024. Advances in Intelligent Systems and Computing*, 2024.

R. Roussel, A. L. Edelen, T. Boltz, D. Kennedy, Z. Zhang, F. Ji, X. Huang, D. Ratner, A. Santamaria Garcia, C. Xu, J. Kaiser, A. Ferran Pousa, A. Eichler, J. O. Lübsen, N. M. Isenberg, Y. Gao, N. Kuklev, J. Martinez, B. Mustapha, V. Kain, C. Mayes, W. Lin, S. Maria Liuzzo, J. St. John, M. J. V. Streeter, R. Lehe and W. Neiswanger. "**Bayesian Optimization Algorithms for Accelerator Physics**," in *Physical Review Accelerators and Beams*, 2024.

J. Kaiser, C. Xu, A. Eichler, A. Santamaria Garcia, O. Stein, E. Bründermann, W. Kuropka, H. Dinter, F. Mayet, T. Vinatier, F. Burkart and H. Schlarb. "**Reinforcement Learning-trained Optimisers and Bayesian Optimisation for Online Particle Accelerator Tuning,**" in *Scientific Reports*, 2024.

J. Kaiser, C. Xu, A. Eichler and A. Santamaria Garica. "**Bridging the Gap Between Machine Learning and Particle Accelerator Physics with High-Speed, Differentiable Simulations**," in *Physical Review Accelerators and Beams*, 2024.

S. Hirlaender, L. Lamminger, S. Pochaba, J. Kaiser, C. Xu, A. Santamaria Garcia, L. Scomparin and V. Kain. "**Towards Few-Shot Reinforcement Learning in Particle Accelerator Control**," in *15th International Particle Accelerator Conference (IPAC)*, 2024.

S. Tomin, J. Kaiser, N. M. Lockmann, T. Wohlenberg and I. Zagorodnov. "**Undulator Linear Taper Control at the European X-Ray Free-Electron Laser Facility**," in *Physical Review Accelerators and Beams*, 2024.

C. Xu, J. Kaiser, E. Bründermann, A. Eichler, A.-S. Müller and A. Santamaria Garcia. "**Beam Trajectory Control with Lattice-agnostic Reinforcement Learning**," in *Proceedings of the 14th International Particle Accelerator Conference (IPAC)*, 2023.

J. Kaiser, A. Eichler, S. Tomin and Z. Zhu. "**Machine Learning for Combined Scalar and Spectral Longitudinal Phase Space Reconstruction,**" in *Proceedings of the 14th International Particle Accelerator Conference (IPAC)*, 2023.

J. Kaiser, O. Stein and A. Eichler, "**Learning-based optimisation of particle accelerators under partial observability without real-world training**," in *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.

F. H. Bahnsen, J. Kaiser, and G. Fey, "**Designing recurrent neural networks for monitoring embedded devices**," in *2021 IEEE European Test Symposium (ETS)*, 2021.

J. Kaiser, K. Bavendiek and S. Schupp, "**Do we need real data? - Testing and Training Algorithms with Artificial Geolocation Data**," in *INFORMATIK 2019: 50 Jahre Gesellschaft für Informatik–Informatik für Gesellschaft*, 2019.

SELECTED TALKS & TUTORIALS

ChatGPT, steuere mir meinen Teilchenbeschleuniger at *Hamburg Innovation Summit*. Hamburg, Germany. 2025.

Differentiable Simulations at *5th ICFA Beam Dynamics Mini-Workshop on Machine Learning for Particle Accelerators*. CERN, Geneva, Switzerland. 2025.

Cheetah: Bridging the Gap Between Machine Learning and Particle Accelerator Physics with High-Speed, Differentiable Simulations at *Fourth MODE Workshop on Differentiable Programming for Experiment Design*. Valencia, Spain. 2024.

Applying Reinforcement Learning to Particle Accelerators: An Introduction at *4th ICFA Beam Dynamics Mini-Workshop on Machine Learning Applications for Particle Accelerators*. Gyeongju, South Korea. 2024.

Large Language Models for Particle Accelerator Tuning at *1st Large Language Models in Physics Symposium (LIPS)*. Hamburg, Germany. 2024.

Reinforcement Learning Tutorial: Application to an Accelerator Problem at *1st Collaboration Workshop on Reinforcement Learning for Autonomous Accelerators (RL4AA'23)*. KIT, Karlsruhe, Germany. 2023.

Reinforcement Learning for the Optimization of Particle Accelerators at *Machine Learning in Engineering Summer School*. Hamburg, Germany. 2022.

Learning-based Optimisation of Particle Accelerators Under Partial Observability Without Real-World Training at *39th International Conference on Machine Learning (ICML)*. Baltimore, Maryland, USA. 2022.

SKILLS

Programming & Tools: e.g. Python, C, C++, Java, MATLAB, Git, Visual Studio Code, Jupyter, Anaconda, PyTest, etc.

Machine Learning: e.g. PyTorch, Stable Baselines3, Weights&Biases, Optuna, Lightning, Scikit-learn, etc.

High Performance Computing: e.g. SLURM

Languages: German (native), English (full professional proficiency), Spanish (basic)

PERSONAL PROJECTS

8mm Film Scanner

Conversion of an old projector to a Regular 8 and Super 8 film scanner, including hardware, control software, a web app, post-processing workflow and documentation.

URL: https://github.com/jank324/8mm-film-scanner

Shutter Speed Tester

Arduino project to help measure and adjust the shutter speeds of analogue cameras.

URL: https://github.com/jank324/shutter-speed-tester